

ONE-WAVE FRAMEWORK

Book 5 — Macro

Chapter 4: Black Holes, Quasars, and White Energy / Cosmic Mirror Gate

Version: 1.0 (draft) Date: July 17, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: A-115 Unified Compression Field, C-301 Mirror Gate, C-309 Friction Limit, B-209 Break Condition, A-112a Traveling Lattice Rupture, Book 5 Ch3 (Supernovae), E-522 Cellular-Stellar Scale Invariance Status: YELLOW (structure, real math inherited) / YELLOW (cosmological application unverified)

Grounding note: this chapter develops something already real, not new — C-301 Mirror Gate has listed “Cosmological level: black hole -> white hole transition” as one of its scale-invariant applications since before this session’s audit began. It was never elaborated into a chapter. This is that elaboration, not a fresh invention.

Gray — Standard Model Reference

A black hole is a region of spacetime where gravity is so strong that nothing, including light, can escape once past the event horizon. Hawking radiation theoretically allows black holes to slowly evaporate over cosmological timescales. A white hole is a time-reversed solution to the same general-relativistic equations — nothing can enter, only exit — but no white hole has ever been observed, and whether they can exist physically (versus being a mathematical artifact of the equations) is unresolved.

A quasar is an extremely luminous active galactic nucleus, powered by matter accreting onto a supermassive black hole — the accretion disk itself, not the black hole directly, produces the enormous luminosity and relativistic jets observed.

Standard Model strength: black hole physics (event horizons, accretion disks, jets) is well-modeled and observationally confirmed (black hole imaging, gravitational wave detections from mergers). Standard Model limitation: white holes remain purely theoretical, and whether any real astrophysical process corresponds to one is unresolved. Quasar jet mechanics have open questions regarding exact launching mechanism.

2D One-Wave Interpretation

In 2D, a black hole is a tear in the lattice — a region where the Friction Limit (C-309) itself breaks down, not merely a region of extreme compression.

This is a stronger and more specific claim than “gravity is very strong here.” C-309 establishes light as a mode that always travels at exactly v_{max} . If a region of the lattice cannot support propagation at all — a genuine structural discontinuity, not just an extreme gradient — light entering that region has nowhere to propagate to. That is a proposed local failure of the propagation medium itself. It has no connection to the Mass-Effect mechanism in C-318, which remains the four-interaction response of a stable recurrence.

A quasar, in this reading, is what a Mirror Gate crossing (C-301) looks like at this scale: the compressed state on one side of the tear undergoes the same symplectic rotation already established for spin-half (C-308) and proposed for the electron/proton system, and re-expresses on the other side as outward ejection.

3D One-Wave Interpretation

In 3D: a black hole is a region where the lattice's structural continuity fails — not simply high beta (coupling) or high gamma (damping), but a genuine tear, meaning C-309's $v_{\max} = \sqrt{\beta_{\max}} * \Delta x / \Delta t$ has no defined value because Δx 's normal lattice-step structure doesn't hold there. This is a candidate definition, not yet distinguished mathematically from “extremely high compression” — flagged directly in Yellow Audit below.

C-301's real Mirror math, applied here directly: $M(\psi_C, \psi_E) \rightarrow (\psi_E, -\psi_C)$, with $M^2 = -I$, $M^4 = I$.

The claim: matter/field-state falling into a black hole undergoes this same Mirror crossing — already established at quantum (spin-half), atomic (shell transition), and biological (nerve relay) scale, per C-301's own pre-existing list. A quasar's relativistic jet is the $M^4=I$ closure completing at cosmological scale: the compressed state re-expresses as ejection on “the other side” of the crossing, matching C-301's own literal wording (“black hole $\rightarrow$ white hole transition”) rather than inventing new vocabulary for it.

Recycling, connected honestly to Book 5 Ch3's Break Condition framing: this IS the same “seeding” mechanism as Ch3's supernova Reset outcome, at a different point in the same family of mechanisms — B-209's Break Condition describes a paired loop failing and releasing material; C-301's Mirror Gate describes what happens to a compressed state AFTER a sufficiently extreme break, specifically the flip-and-reappear structure. These may be the same underlying event viewed at two different stages (approach-to-break vs. the crossing itself) — a real, checkable candidate relationship, not yet verified term-by-term.

Mathematics

Mirror operator (C-301, real, inherited unmodified): $M = [[0,1],[-1,0]]$, $M^2 = -I$, $M^4 = I$

Friction limit (C-309, real, inherited unmodified): $c = v_{\max} = \sqrt{\beta_{\max}} * \Delta x / \Delta t$

Candidate black hole condition (new, NOT derived — a proposed definition requiring further work): A black hole exists where Δx 's normal lattice-step definition fails, making $v_{\max}$ undefined rather than merely very small. This is different from a Persistent Mode with extremely high compression (which would still have a defined, if tiny, $v_{\max}$) — the distinction between “very strong compression” and “structural failure” has not been made mathematically precise here.

Candidate quasar condition (new, NOT derived): Jet luminosity/velocity as a function of the $M^4=I$ closure completing — no functional form proposed, only the qualitative claim that the closure produces the ejection.

Yellow relocation model for a moving tear

A-112a now supplies a minimal conservative model for how a localized tear could move without leaving an indefinitely growing permanent fracture. Let r_i^n be normalized rupture occupancy at lattice site i and step n . For right-moving transport:

$$\begin{aligned} r_i^{(n+1)} &= r_i^n - \nu(r_i^n - r_{(i-1)}^n) \\ \nu &= v \Delta t / \Delta x \\ 0 &\leq \nu \leq 1. \end{aligned}$$

Under periodic or zero-flux boundaries:

$$\begin{aligned} R_n &= \sum_i r_i^n \\ R_{(n+1)} &= R_n. \end{aligned}$$

The flux proof is telescoping: what leaves one site enters the next. Opening and reclosure are the positive and negative parts of the same state change:

$$\begin{aligned} O_i^n &= [r_i^{(n+1)} - r_i^n]_+ \\ C_i^n &= [r_i^n - r_i^{(n+1)}]_+ \\ \sum_i O_i^n &= \sum_i C_i^n. \end{aligned}$$

This formalizes “open ahead <-> close behind.” It does not yet derive ν from black-hole pressure, gravity, or spacetime observations.

Permanent scar is a separate variable d_i , not an automatic result of rupture transport:

$$d_i^{(n+1)} = \text{clip}(d_i^n + \kappa_d[r_i^n - r_{\text{damage}}]_+ (1 - d_i^n) - \lambda_d d_i^n, 0, 1).$$

A normal traveling defect stays below the damage threshold or repairs behind itself. A propagating fracture is the different regime in which scar length grows. The mathematical distinction now exists; the claim that an astrophysical black hole is such a defect remains unverified.

White Energy: Population-Scale Return

One-Wave rejects expansion of space. The term **White Energy** names the outward-return side of the cosmic circulation: stored compression is released through quasar/white-hole-scale Mirror-Gate events and reinjected into the field.

The “massive ATP ejection” phrase is an architectural analogy. It means stored energy is released to drive the next system state. It does not claim astrophysical objects use cellular chemistry.

The compact reservoir follows E-530:

$$\frac{dU_C}{dt} = P_{\text{cap}} + P_\nu - \lambda_C U_C - D_W h U_C,$$

$$P_W = D_W h U_C.$$

The release state turns on at U_{on} and off at $U_{\text{off}} < U_{\text{on}}$. Recharge, state-dependent discharge, and hysteresis are all required for repeated events.

White Energy remains part of the energy inventory while it travels outward:

$$\frac{dE_W}{dt} = P_W - P_{W \rightarrow \chi} - \Phi_{W, \partial \Omega}.$$

For a closed accounting domain,

$$E_{\text{tot}} = E_\gamma + E_\chi + E_\nu + E_C + E_W, \quad \frac{dE_{\text{tot}}}{dt} = 0.$$

Static long-term balance requires

$$\langle P_W \rangle = \langle P_{\text{cap}} + P_\nu - \lambda_C U_C \rangle.$$

No Hubble parameter, scale factor, or negative-pressure equation belongs in the One-Wave model.

Redshift and Return Loop

E-528 gives the static propagation law

$$1 + z = \exp \left(\int_0^D \kappa_\gamma d\ell \right).$$

Energy removed from a Propagating Light Mode enters the compression field. E-529 proposes Low-Coupling Return Modes as one transport channel toward compact reservoirs. E-530 closes the candidate loop:

compression and Mass Effect

- > gravity / extended compression
- > tired-light redshift and field transfer
- > Low-Coupling Return Modes
- > compact reservoir
- > Mirror-Gate threshold
- > White Energy ejection
- > field reinjection

Yellow Audit

- White Energy is a naming and architecture lock, not experimental closure.
- The quasar/white-hole identification remains Green.
- The tired-light coefficient is not derived.
- Neutrino deposition into compact reservoirs is not derived.
- Population-scale static energy balance has not been simulated.
- No expansion variable is permitted in future One-Wave equations.

Future Work

1. Simulate E-528 through E-530 as one closed energy-conserving system.
2. Calibrate a static redshift coefficient without image blurring or unexplained heating.
3. Derive Return-Mode production and deposition coefficients.
4. Map compact-reservoir parameters to quasar observations without reverse fitting.